

Newsense: A Database Application Leveraging User Input to Analyze the Reliability of News Sources

Jake Leigh, Maverick Donald-Wright, Ngoc (Marie) Le
Northeastern University, Boston, MA, USA

Abstract

The goal of this project is to test a Database application, *Newsense*, by populating the database with sample data and testing various user cases. *Newsense* serves as a resource for individuals who may be overwhelmed by the number of news outlets on air or unsure of which outlets are generally considered reliable.

Therefore, the app features various user input options including written reviews and a rating system on a scale from 1 to 5 - with users optionally providing their self-reported political-affiliation, country, and region (state, province, etc.). These reviews are for articles, rather than specific news outlets, and each user may submit only one review per article. Reviewing articles also allows users to see which articles have received the most or highest ratings. Further, there is a credibility rating system in place, allowing users to rate other user profiles. Since the app relies on user input for reviews and self-reported information, credibility ratings allow users to discern whether reviews by a given user are reliable.

Introduction

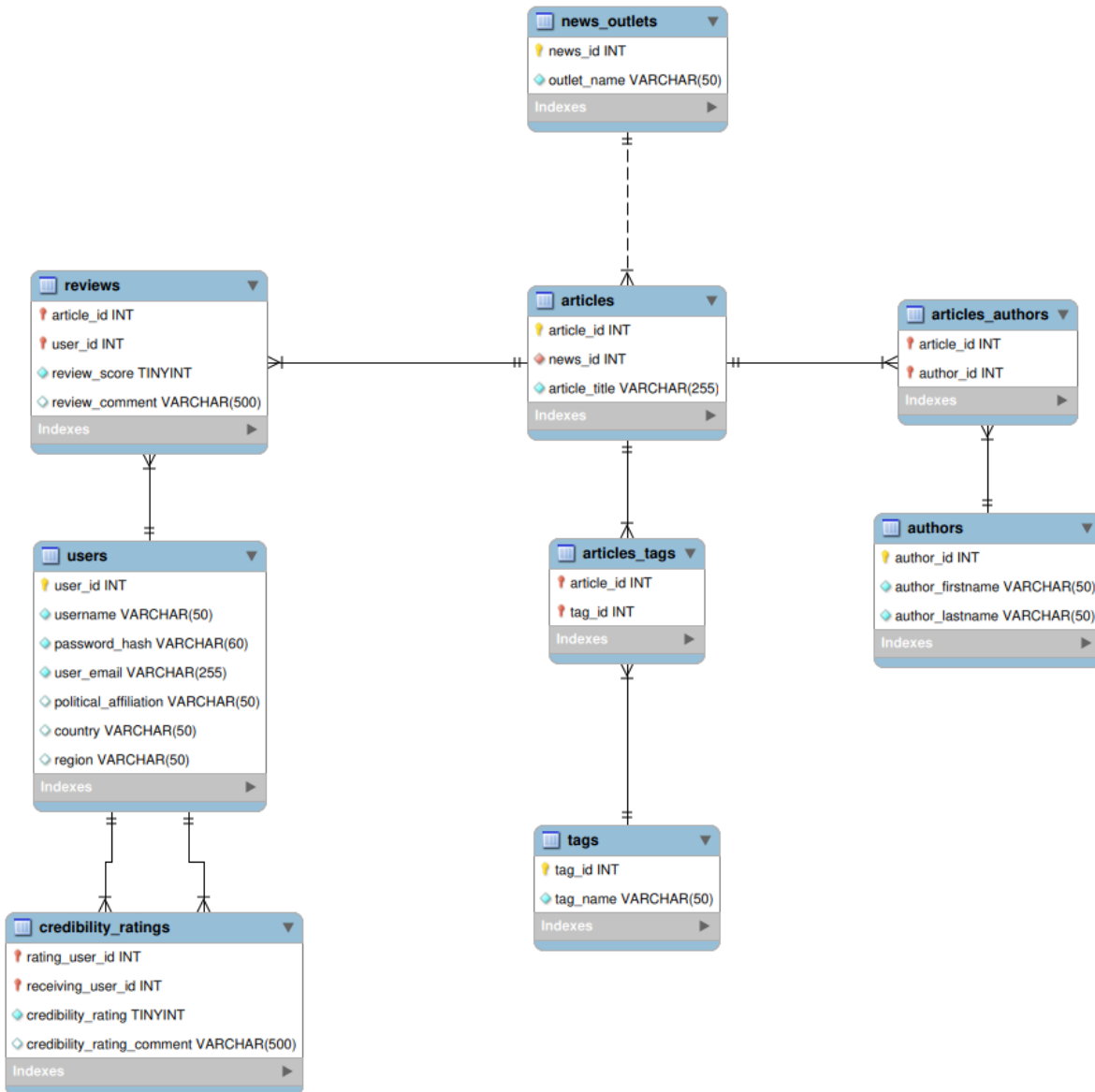
NewSense is a browser extension application that seeks to capture a diverse array of opinions in order to quantify the trustworthiness of news sources while accounting for user biases. To achieve this, when a user (denoted as user1 for example) clicks on a news article, they can submit a trust review based on their own reading of the article. If the article is not already in the app's database, the article is added to the "articles" table. Another user (denoted as user2) can then submit a credibility rating against user1. The app can then query the database to collect summary statistics (i.e. unweighted averages, weighted averages based on user credibility) on news outlets in order to compile data on how trustworthy a given news outlet or individual article is.

NewSense is designed to provide users with a reliable system for verifying and assessing the credibility of online news sources. Its significance can be outlined as follows:

- **Provides a Centralized Database for News Credibility:**
Offers a structured and accessible database that stores, organizes, and manages credibility information for various news outlets.
- **Serves as a Useful Tool for Different Users:**
Beneficial for students, journalists, researchers, and the general public who seek verified and trustworthy sources of news information.
- **Encourages Innovation in Information Management:**

Integrates database management with analytical features, promoting innovative solutions in data-driven media evaluation.

Database Design



Tables:

- **news_outlets (pk news_id)**: Keeps track of news outlets by name
- **articles (pk article_id)**: Keeps track of news articles by title. Each article is associated with a news outlet via **fk news_id**
- **articles_authors (pk+fk article_id, author_id)**: Join table between **articles** and **authors**
- **authors (pk author_id)**: Keeps track of authors by name

- **articles_tags (pk+fk article_id, tag_id):** Join table between **articles** and **tags**
- **reviews (pk+fk article_id, user_id):** Enables a many-to-many relationship so that users can leave reviews on multiple articles (pk uniqueness prevents users from reviewing an article twice)
- **users (pk user_id):** Keeps track of user identity, login info, and metadata
- **credibility_ratings (pk+fk rating_user_id, receiving_user_id):** Enables a self-referential many-to-many relationship between users, representing users to rate other users based on perceived credibility

Relationships:

- **articles↔authors (many-to-many):** Facilitated via the **articles_authors** join table. Each article can have multiple authors, and each author can have written multiple articles
- **articles↔tags (many-to-many):** Facilitated via the **articles_tags** join table. Each article can have multiple tags, and each tag can be chosen on multiple articles
- **news_outlets→articles (one-to-many):** Each news outlet can publish multiple articles, but each article is associated with one news outlet. **news_outlets** is the parent, **articles** is the child
- **users→reviews→articles (two one-to-many relationships that form a many-to-many relationship):**
 - **users→reviews (one-to-many):** Each user can write multiple reviews, but each review is associated with one user
 - **articles→reviews (one-to-many):** Each article can have multiple reviews, but each review is posted on one article
 - **users→articles (many-to-many):** If treating **reviews** as a join table, users can review multiple articles, and articles can be reviewed by multiple users
- **users→users (self-referential many-to-many):** Through **credibility_ratings** as a join table, each user can leave credibility ratings on multiple other users, and each user can receive multiple credibility ratings from other users. Each user→user pair must be unique, though, so one user cannot form multiple ratings for a single other user.

Data Source/Methods

We manually created dummy data to demonstrate the capabilities of the database at a small scale. It is easily reproducible by running the populate script

Generation:

- News outlets used are a selection of real news outlets
- News articles are fabricated, but populated with realistic article names
- Review, tag, user political affiliation, and rating data may not represent real-world distributions, but was chosen to create meaningful variation for demonstrating the capabilities of the database

Assumptions:

- The application, not the database, constrains the review scores and credibility ratings to a 1-5 scale.
- The application would use a stronger hashing algorithm. The dummy data uses md5 hashes
- The application will ensure that users must enter a country first if they want to enter a region for their profile

User Case Analyses

1. **Question:** What is the average rating for each news outlet?

Summary: For users seeking to compare the reliability of different news sources, what is the mean trustworthiness rating for each individual outlet in our database?

```

4  -- What is the average rating for each news outlet?
5
6  • SELECT outlet_name, ROUND(AVG(review_score),2) AS "avg_rating"
7    FROM news_outlets
8         LEFT JOIN articles USING (news_id)
9         JOIN reviews USING (article_id)
10   GROUP BY news_id
11   ORDER BY avg_rating DESC;
12

```

100% 20:17

Result Grid Filter Rows: Search Export:

outlet_name	avg_rating
The Huntington News	4.43
The Washington Post	4.00
WIRED Magazine	3.50
The New York Times	3.00

2. **Question:** What is the average reliability rating across all news outlets?

Summary: For users looking to understand the overall trustworthiness of news sources, what is the typical rating when considering all outlets in our database?

```

13  -- What is the average rating across all news outlets?
14
15  • SELECT ROUND(AVG(avg_rating),2) AS "avg_overall_rating"
16    FROM (SELECT outlet_name, AVG(review_score) AS "avg_rating"
17          FROM news_outlets
18               LEFT JOIN articles USING (news_id)
19               JOIN reviews USING (article_id)
20             GROUP BY news_id
21             ORDER BY avg_rating DESC) AS outlet_ratings;
22

```

100% 26:11

Result Grid Filter Rows: Search Export:

avg_overall_rati...
3.73

3. **Question:** How do credibility-weighted ratings compare to unweighted average ratings?

Summary: For users interested in more accurate assessments, how does the average rating of news outlets change when we factor in the credibility of the users providing the ratings, compared to the simple overall average?

```

31 -- If we weigh the ratings based on user credibility, how does this compare to true average ratings?
32 SELECT
33     n.outlet_name,
34     a.article_title,
35     ROUND(AVG(r.review_score), 2) AS unweighted_avg_rating,
36     ROUND(SUM(r.review_score * IFNULL(c.avg_credibility, 1)) / SUM(IFNULL(c.avg_credibility, 1)), 2) AS weighted_avg_rating
37 FROM news_outlets n
38 JOIN articles a ON n.news_id = a.news_id
39 JOIN reviews r ON a.article_id = r.article_id
40 JOIN (
41     SELECT receiving_user_id, AVG(credibility_rating) AS avg_credibility
42     FROM credibility_ratings
43     GROUP BY receiving_user_id
44 ) c ON r.user_id = c.receiving_user_id
45 GROUP BY a.article_id
46 ORDER BY weighted_avg_rating DESC;
47

```

outlet_name	article_title	unweighted_avg_rati...	weighted_avg_rati...
WIRED Magazine	Your Smart Devices Leak More Than You Think...	5.00	5.00
The Huntington News	About the Northeastern Community Dogs: Coop...	4.67	4.68
The Huntington News	How Co-op's Shape Northeastern Students' Lives	4.33	4.22
The Huntington News	What Caused the AWS Incident? A Northeaster...	4.00	4.00
The Washington Post	More states are offering cheap health plans to f...	4.00	4.00
The New York Times	Zohran Mamdani Elected Mayor: What This Me...	3.00	3.00
WIRED Magazine	Silicon Prices Are Rising. Here's How That Impa...	2.00	2.00

4. **Question:** What is the average review score for an article with each tag?

Summary: For users interested in how specific topics or categories are perceived, what is the typical reliability rating for articles associated with each tag in our database?

```

48 -- What is the average review score for an article with each tag?
49 SELECT t.tag_name, AVG(r.review_score) AS avg_rating
50 FROM tags t
51 JOIN articles_tags ajt ON t.tag_id = ajt.tag_id
52 JOIN articles a ON ajt.article_id = a.article_id
53 JOIN reviews r ON a.article_id = r.article_id
54 GROUP BY t.tag_id
55 ORDER BY avg_rating DESC;
56

```

tag_name	avg_rating
Dog	4.6667
Northeastern	4.4286
Econ	3.8000
Tech	3.6667
Politics	3.5000

5. **Question:** Which authors tend to be more well received (average score for articles written by each author, ordered desc)?

a. How might the number of published articles impact an author's average score?

Summary: For users looking to identify reliable contributors, which authors have the highest average scores for the articles they have written, ranked from highest to lowest? Also, for users interested in understanding author performance, does the quantity of articles an author has published influence their average article score, and if so, how?

```

57 -- Which authors tend to be more well received (average score for articles written by each author, ordered desc)?
58 -- How might the number of published articles impact an author's average score?
59 • SELECT au.author_firstname, au.author_lastname, COUNT(DISTINCT ar.article_id) AS num_articles, ROUND(AVG(r.review_score), 2) AS avg_rating
60 FROM authors au
61 JOIN articles_authors arjau ON au.author_id = arjau.author_id
62 JOIN articles ar ON arjau.article_id = ar.article_id
63 JOIN reviews r ON ar.article_id = r.article_id
64 GROUP BY au.author_id
65 ORDER BY avg_rating DESC;
66

```

author_firstname	author_lastname	num_articles	avg_rating
Paws	Husky	2	4.50
Noah	ILoveNortheasternToAnUnhealthyDegree	2	4.25
Riley	Econ	1	4.00
Eliza	Tech	2	3.50
Jimothy	Politics	1	3.00

6. Question: How does Political Affiliation count compare to Region count?

- How many users identify with each political affiliation?
- What is the count of users for each region in the database?

Summary: For users interested in understanding the distribution of our user base, how many users identify with each political affiliation, how many are from each region, and how do these counts compare overall?

```

23 -- How does Political Affiliation count compare to Region count?
24
25 • SELECT region, political_affiliation, COUNT(political_affiliation) as "num_affiliation"
26 FROM users
27 GROUP BY political_affiliation, region;
28

```

region	political_affiliati...	num_affiliati...
New York	Democrat	2
Puerto Rico	Libertarian	1
Ohio	Republican	1

7. Question: Which tags are most commonly associated with articles from a specific news outlet?

Summary: For users looking to understand the main topics covered by a particular news source, which article tags appear most frequently in that outlet's content, helping users identify the focus areas and themes emphasized by that news provider?

```

68 -- Which tags are most commonly associated with articles from a specific news outlet?
69 • select c.outlet_name, c.tag_name, c.tag_used
70 from (
71     select n.outlet_name, t.tag_name, count(*) as tag_used
72     from tags t
73     join articles_tags at on at.tag_id = t.tag_id
74     join articles a on at.article_id = a.article_id
75     join news_outlets n on n.news_id = a.news_id
76     group by n.outlet_name, t.tag_name
77 ) c
78 join (
79     select outlet_name, max(tag_used) as max_tag_used
80     from (
81         select n.outlet_name, t.tag_name, count(*) as tag_used
82         from tags t
83         join articles_tags at on at.tag_id = t.tag_id
84         join articles a on at.article_id = a.article_id
85         join news_outlets n on n.news_id = a.news_id
86         group by n.outlet_name, t.tag_name
87     ) x
88     group by outlet_name
89 ) m
90 on c.outlet_name = m.outlet_name
91 and c.tag_used = m.max_tag_used
92 order by c.outlet_name;
93

```

outlet_name	tag_name	tag_used
The Huntington News	Northeastern	3
The New York Times	Politics	1
The Washington Post	Politics	1
The Washington Post	Econ	1
WIRED Magazine	Tech	2

8. **Question:** Which news outlets have the most ratings (top 5)?

Summary: For users interested in understanding which news sources are most actively evaluated by our database users, which five outlets have received the highest number of ratings, indicating where user engagement is greatest?

```

94 -- Which news outlets have the most ratings (top 5)?
95 • select count(review_score) as review_ratings, outlet_name
96 from reviews r
97 join articles a on a.article_id = r.article_id
98 join news_outlets n on n.news_id = a.news_id
99 group by n.outlet_name
100 order by review_ratings desc
101 limit 5;
102

```

review...	outlet_name
7	The Huntington News
2	WIRED Magazine
1	The New York Times
1	The Washington Post

9. **Question:** Do certain news outlets receive systematically higher or lower credibility ratings for the same tags or topics?

Summary: For users interested in comparing news sources on a topic-by-topic basis, are there patterns in which specific outlets consistently receive higher or lower reliability ratings for articles sharing the same tags, helping users identify potential biases or trends across sources?

```

103 -- Do certain news outlets receive systematically higher or lower credibility ratings for the same tags or topics?
104 select tag_name, n.outlet_name, round(avg(r.review_score), 2) as avg_rating, count(*) as num_reviews
105 from reviews r
106 join articles a on r.article_id = a.article_id
107 join news_outlets n on n.news_id = a.news_id
108 join articles_tags at on at.article_id = a.article_id
109 join tags t on t.tag_id = at.tag_id
110 group by t.tag_name, n.outlet_name
111 order by t.tag_name, avg_rating desc;
112

```

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Result Grid Filter Rows: Search Export:

review...	outlet_name
7	The Huntington News
2	WIRED Magazine
1	The New York Times
1	The Washington Post

10. Question 11: What are the counts of user political affiliation for users who submitted ratings for each news outlet?

Summary: For users interested in understanding how political perspectives relate to news evaluations, how many users of each political affiliation have submitted ratings for each news outlet, helping reveal patterns in user engagement and potential biases in source ratings?

```

113 -- What are the counts of user political affiliation for users who submitted ratings for each news outlet?
114 select n.outlet_name, u.political_affiliation, count(*) as affiliation_count
115 from reviews r
116 join users u on u.user_id = r.user_id
117 join articles a on a.article_id = r.article_id
118 join news_outlets n on n.news_id = a.news_id
119 group by n.outlet_name, u.political_affiliation
120 order by n.outlet_name, affiliation_count desc;
121

```

47:106

Result Grid Filter Rows: Search Export:

outlet_name	political_affiliati...	affiliation_co...
The Huntington News	Democrat	4
The Huntington News	Libertarian	2
The Huntington News	Republican	1
The New York Times	Democrat	1
The Washington Post	Libertarian	1
WIRED Magazine	Democrat	1
WIRED Magazine	Republican	1

Conclusions

The results of our project were somewhat promising with test data, but our sample queries revealed potential flaws with *Newsense*. The database does seem to comprehensively cover potential areas of interest for users such as written and numerical reviews of articles, article tags, other user demographics and credibility ratings. However, the inherent bias of the application was also shown through the use of these features in the sample queries.

While user credibility ratings are intended to better eliminate the subjectiveness of a user_based rating system, users may not rate each other based on written reviews of articles, but on seemingly unrelated aspects of their profile - such as username or region. Further, if there are a significant number of users from one region or with a particular political affiliation, then the reviews could be skewed towards news outlets of the region or news outlets with more reviewers aligning with that political affiliation. The sample data, for example, contains more democrats, thus outlets typically favored by democrats may receive more and likely higher ratings. As the

user base expands, the use case for Question 11 will be helpful in seeing the distribution of different political affiliations of users who reviewed each outlet and be furthered by also showing the average rating for each political group for each news outlet.

The method of determining news outlet ratings may need to include outlet-specific ratings, rather than relying on article ratings. A news outlet may receive worse ratings for a single controversial or unpopular article. More significantly, the reviews of articles may be based on a user's opinion of the author(s), which should not be weighted in a review of a news outlet. There is still merit to an article-based rating system, however, as the outlets may be rated primarily on the basis of content.

Author Contributions

While all of us brought something meaningful to the table, each of us contributed to the project in different ways. Maverick played a key role early on by setting up the database structure from our proposal and writing the necessary queries to create the tables. Jake helped refine the database design early on before writing insert queries in order to populate the tables with example data for use in application-relevant queries. Ngoc focused on summarizing our work for the slideshow, creating a clear presentation that covered our introduction, database design, research questions, main analyses, and key takeaways. Alongside these individual roles, we all collaborated on the written report, dividing the sections among us and merging them into a unified final document. After several iterations and refinements, we were able to build the tables, craft effective queries, and deliver an implementation that reflected our overall vision for the project.